



MANIPAL

ACADEMY of HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5101 Cryptology Theory – 2

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Assignment - 4

Decrypt the following ciphertext using the hill cipher.

① Cipher Text : HLDTPIHJDXWQCMAG

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Use (A=0, B=1, ..., Z=25)

Find the plain text

Step 1: Find the determinant of K:

$$\det(K) = 3(7) - 5(2) = 21 - 10 = 11$$

Step 2: Find K^{-1}

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$11^{-1} \pmod{26} \quad 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 - 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$11(2) + 4 + (4(2) + 3)(-3) + (3(1) + 1)(2) = 1$$

$$11(2) + 4 + 4(-1) + 3(-3) + 3(2) + 1(2) = 1$$

$$(4(2)+3)2 + 3(1)+1 + (3(1)+1)(-1) + 3(-1)+1(2) = 1$$

$$4(4) + \cancel{3(2)} + 3(3) + 1 + 3(-1) + 1(-1) + 3(-1) + 1(2) = 1$$

$$(3(1)+1)4 + 3(3) + 1 + 3(-2) + 1(1) = 1$$

$$3(4) + 1(4) + 3(3) + 1(2) + 3(-2) = 1$$

$$3(5) + 1(6) = 1$$

$$\underline{\underline{19}}$$

$$11^{-1} \bmod 26 = 19 \bmod 26 = 19$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

Step 3: Divide Ciphertext into pairs

HL DT QI HJ DX WQ CM AG

Step 3: Divide ciphertext into pairs

HL DT QI HT DX WQ CM AG

Formula: $P = K^{-1}C \pmod{26}$

$$HL = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{matrix} T \\ O \end{matrix}$$

$$DT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 975 \\ 122 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$QI = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{matrix} F \\ C \end{matrix}$$

$$HT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$DX = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 13 \end{bmatrix} \pmod{26} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$WQ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} E \\ S \end{matrix}$$

$$CM = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{matrix} S \\ A \end{matrix}$$

$$AG = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{matrix} G \\ E \end{matrix}$$

Plaintext = TOP SECRET MESSAGE

② Consider the Hill Cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} \text{ Ciphertext: RXAODY}$$

Find the plaintext for the above Ciphertext

Step 1: Find determinant of K

$$\det(K) = 24(14) - 19(5) = 336 - 95 = 241$$

$$241 \bmod 26 = 7$$

Step 2: Find K^{-1}

$$K^{-1} = (\det(K))^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$(\det K)^{-1} = 7^{-1} : 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[26 + 7(-3)] + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} \bmod 26 = 15 \bmod 26 = 15$$

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

Step 3: Divide ciphertext into pairs:

RX AO DY. Formula: $P = K^{-1} C \pmod{26}$

$$RX = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 59 \\ 539 \end{bmatrix} \pmod{26} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \pmod{26} = \begin{bmatrix} 30 \\ 539 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 19 \end{bmatrix} = \begin{matrix} F \\ R \end{matrix}$$

Plaintext: FLOWER

Confusion vs Diffusion

Analyze the following classical ciphers with respect to Confusion & Diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext.

Cipher	Does it change the character / symbol value?	Does it change the character position?	Confusion / Diffusion	Justification
Shift Cipher	Yes	No	Confusion	Each plaintext character is replaced by another character by shifting its value. The position remains unchanged.
Multiplicative Cipher	Yes	No	Confusion	Each character value is multiplied by a key modulo 26. Character positions remain unchanged.
Affine Cipher	Yes	No	Confusion	Each character is transformed using $C = (aP + b) \bmod 26$. The symbol value changes but its position does not.

Transposition Cipher	No	Yes	Diffusion	The character themselves remain unchanged, but their positions are rearranged

Based on your observations, explain the relationship between substitution and confusion and between transposition and diffusion.

Relationship between substitution and confusion:

- * Substitution ciphers such as shift, multiplicative and affine cipher replace one symbol with another.
- * Therefore, they primarily provide confusion by making the relationship between the plaintext symbols and ciphertext symbols less direct.

Relationship between transposition & diffusion

- * A transposition cipher does not change the actual characters. Instead, it rearranges their positions.
- * This spreads the plaintext characters across different positions in the ciphertext and therefore provides diffusion.

Substitution → Confusion

Transposition → Diffusion

To ans position

Encryption

Encrypt the message "The Arrow is pointing north direction"

Using

- simple with key = 4
→ Key = 4

No of Columns = 4

T	H	E	A
R	R	O	W
T	S	P	O
I	N	T	I
N	G	N	O
R	T	H	D
T	R	E	C
T	I	O	N

Ciphertext = TRIINRITHRSNGTRIEOPTNHEO
AWOIODCN

Keyword with keyword = ATLAS

Keyword with keyword = ATLAS

Keyword length = 5, $\therefore$ No. of column = 5

A	T	L	A	S
T	H	E	A	R
R	O	W	I	S
P	O	T	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

Ciphertext: TRPIRRO AINNDIX EWTGHCH
RSTOIX HOONTEN

- double

key1: BREAK, Key2: log

→ Key2: ~~log~~ BREAK, length = 5, No. of column = 5

B	R	E	A	K
T	H	E	A	K
R	O	W	I	S
P	O	T	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

Initial ciphertext = AINNDIXTRPIRROEWIGHCHX
XHOONTEN

→ Key = LOG, Length = 3, No. of Columns = 3

L	O	G
A	I	N
N	D	T
X	T	R
P	I	R
R	O	E
W	I	G
H	C	X
R	S	T
O	I	I
X	H	O
O	N	T
E	N	X

Final Ciphertext is

NTRREGXTIOTX ANXPRWHROXOE
IDTIOI CSIHNN

Decryption

- Simple with key 10

TIDIE MRNME REOGO SANOW RSLTP
 EEEEO SSSNU AHTUD BTENP GODAE
 PETXD ELNSX

→ String length = 60, key = 10

$$\frac{60}{10} = 6, \text{ No. of rows} = 6$$

T	R	O	O	P	S	H	E	A	D
T	N	G	W	E	S	T	N	E	E
D	M	O	R	E	S	U	P	P	L
I	E	S	S	E	N	D	G	E	N
E	R	A	L	D	U	B	O	I	S
M	E	N	T	O	A	I	D	X	X

Plaintext: TROOPS HEADING WEST NEED
 MORE SUPPLIES SEND GENERAL
 DUBOIS MEN TO AID

- keyword

SECRET

Ciphertext: TLEPT NHPTR SDIIT MAXSA
XERXT ANDNI SNFOT X

Key = SECRET, Key length = 6, No. of Columns = 6

Ciphertext length = 36, No. of rows = $\frac{36}{6} = 6$

SECRET

5 2 1 4 3 6

→ Numbering of the keyword.

S	E	C	R	E	T
T	H	T	S	I	S
A	P	L	A	I	N
N	T	E	X	T	F
D	R	D	E	M	O
N	S	T	R	A	T
I	D	N	X	X	X

Plaintext: THIS IS A PLAIN TEXT FOR
DEMONSTRATION.

- Keyword

Keyword - PLAIN

CIPHER - ~~VERY~~ VEY EAX ARA ATX HGD

- Keyword = PLAIN, length = 5, No. of columns = 5

Ciphertext Length = 15, No of rows = $\frac{15}{5} = 3$

	P	L	A	I	N
	H	A	V	E	A
	G	R	E	A	T
	Q	A	Y	X	X

Plaintext is

HAVE A GREAT DAY

Permutation

Find the inverse permutation of for.

① 4 3 1 5 2

Position	1	2	3	4	5
P-box	4	3	1	5	2
Inverse Permutation	3	5	2	1	4

Inverse Permutation, $P^{-1} = (3, 5, 2, 1, 4)$

② 7 4 1 3 5 2 6

Position	1	2	3	4	5	6	7
P-box	7	4	1	3	5	2	6
Inverse Permutation	3	6	4	2	5	7	1

Inverse Permutation, $P^{-1} = (3, 6, 4, 2, 5, 7, 1)$

③ 4 1 3 2

Position	1	2	3	4
P-box	4	1	3	2
Inverse Permutation	2	4	3	1

Inverse Permutation, $P^{-1} = (2, 4, 3, 1)$

4. 3 5 1 4 2

Position	1	2	3	4	5
P-box	3	5	1	4	2
Inverse Permutation	3	5	1	4	2

Inverse Permutation, $P^{-1} = (3, 5, 1, 4, 2)$